[illegible] progress [illegible] field of all subjects

Evolutionary ideas applied to all, including psych - mental.

11 From Systems to Specialties: The Crucial Half Century (1870–1920)

If we quickly thumb through the preceding chapters, one characteristic amid the diversity of topics and claims stands out and serves to divide all of the past from what is now taken to be the proper study of psychology. That characteristic is easier to detect than to define. Perhaps the term "system" serves most economically. If we ask, for example, what is common to the psychologies of Plato, Aristotle, Aquinas, Hobbes, Descartes, Locke, and the rest, it turns out to be their uniform commitment to develop a *system* of psychology able to embrace the fullest range of psychological phenomena: thought, emotion, memory, conduct, morality, government, aesthetics, and so forth. When we compare this commitment with the activities that most clearly identify contemporary psychology, we find a quite striking difference. To a certain degree, Freudian theory has been applied to a variety of personal and cultural matters and, to a lesser degree, modern behaviorism has attempted to address various aspects of social life and social organization. But neither Freud nor the modern behaviorist would claim that his approach to psychology is designed to cover as broad a spectrum of phenomena as the approach that was routinely considered just a century ago.

The more obvious explanations of this difference are uninforming. It could be argued, for example, that since earlier attempts failed, modern psychologists have learned to be more modest in their goals, more conservative in their speculations. Yet, if we look at the history of any significant intellectual undertaking, the same failures are common, but the surrender is not. Since the time of the pre-Socratics, there have been numerous attempts to account for all physical phenomena through a small set of universal laws, and none of these attempts succeeded. Nonetheless, today's theoretical physicist is just as committed as the ancients to an all-encompassing theory of matter. Similarly, the pages of intellectual history are filled with attempts to provide a faultless theory of politics, but none has surfaced and won the unopposed approval of the human race. Still, every year thoughtful scholars begin to sketch out a new theory, a new set of basic premises, a new "solution." Thus, we cannot explain the difference between the older and the new psychologists simply in terms of a resignation produced by prior failures.

A more subtle explanation begins with a denial of the claim. Modern psychology, on this account, pursues the same objectives that prompted all earlier endeavors, but it begins with more elemental processes and does not advance to complex phenomena until these processes have been understood. What is unsatisfying about this sort of claim is that it generally is moot on the question of which processes must be understood before the larger issues can be addressed. Is it true that contemporary studies of visual perception are undertaken as the preliminary stage of what is intended to be a developed psychology of aesthetics? Does today's psychologist study the principles of learning and memory *so that* the ageless issues of epistemology might be settled? These questions are not posed to depreciate modern efforts, but to weigh the validity of the claim that these efforts have been deliberately chosen in the interest of larger objectives.

Finally, we may take as an illustration of obvious but unenlightening explanations the one that draws a sharp line between science and every other form of inquiry. Thus, modern psychology is the way it is because it is "scientific," whereas earlier psychologies were not. What is troublesome about this account is a two-fold dilemma: First, as Chapter 1 attempted to show, it is far from clear that psychology is "scientific" or is in any demanding respect somehow "more scientific" than the psychological contributions of the past. It is clearly more *experimental,* but this is a rather different matter. Secondly, it is odd to use the adjective "scientific" as a way of explaining why a subject or range of issues has been constricted. That is, from the mere fact or allegation that X is a science, it does not follow that X is addressed to or interested in a narrower range of issues than Y.

There are, of course, other kinds of explanations too frivolous to consider. For example: Today's psychologists are simply not as bright or clever as the geniuses of old; today's psychologists are more practical and less speculative than their ancestors; today's psychologists have simply chosen their problems with more precision and have accepted as problems only those that might be settled experimentally. The first of these explanations is offensive without being documented. The second and third beg the question, rather than answer it.

By way of introducing the present chapter, it is useful to test an explanation less obvious than the foregoing but truer to the history of the trends. We might best begin with the central fact: Today's psychology is dominated by a relatively small number of highly specialized fields of inquiry and practice, and, in large measure, each of these fields has developed and continues to develop independently of the others and is a manner that is often indifferent to the others. Thus, we have (a) the psychology of personality, (b) animal learning and memory, (c) human psychophysics, (d) psychotherapy, (e) social psychology, and (f) genetic psychology. This is scarcely an exhaustive list, but it does illustrate the variety of separate issues occupying today's specialists. Within this list we recognize the relative independence/indifference just noted. The social psychologist can undertake experimental and theoretical programs without ever consulting the facts or methods of psychophysics. The psychophysicist concerned with auditory discrimination need not explore the nuances of psychotherapy. The psychotherapist proceeds without any or at least much help from the literature on comparative psychology. This fact is neither "good" nor "bad," a cause for joy or sorrow, a subject for concern or aloofness. But it is a fact of history, and one to be understood at least partly in historical terms. As previous chapters have shown, it is not a fact that appeared suddenly, but one that was taking shape for centuries. Every one of the old *isms* contributed to it: empiricism, rationalism, idealism, materialism, and nativism. So too did advances in the physical and biological sciences; and so especially did the general divorce between science and philosophy, which was decreed toward the middle of the nineteenth century. At about this time and for an additional fifty years or so, a number of scientists and scholars combined the various *isms* and scientific advances within the context of the great divorce and thereupon created special fields of psychological inquiry and practice. These individuals are the subject of the present chapter, the bridge-builders allowing us to go from the promise of a "scientific psychology" to the realities of contemporary psychology.

Comparative Psychology—From Anthropomorphism to Behaviorism

Throughout the Enlightenment the attacks on Cartesianism often included special criticism of Descartes "automaton" theory of animal psychology. As early as Pierre Bayle's *Dictionary* there were vigorous defenses of the view that all human faculties can be found in one form or another among the other advanced species and that man's claimed uniqueness was little more than vanity.[1] But Franz Joseph Gall framed the issue squarely in terms of evolutionary biology fifty years before Darwin's *Origin of Species.*[2] Gall's accomplishments would become blurred by the subsequent history of his system of phrenology, but his actual place in the history of physiological psychology remains secure. By the standards of his time, he was a painstaking anatomist and a brilliant theorist. Even before the end of the eighteenth century, he had begun research on the fetal and post-natal development of the nervous systems of a number of species, including man. He offered evidence and strong arguments favoring the conclusion that the degree of moral and intellectual prowess displayed by any animal was tied completely to the degree of cerebral development attained by

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the animal. He called for a new kind of taxonomy, one based upon cerebral evolution and functional capacities rather than merely general anatomy.

As has been noted in earlier chapters, the idea of progress was the preoccupation of the eighteenth century. Philosophical expressions of the evolutionary point of view were common, and even scientific versions were not rare. By the early nineteenth century, this same point of view had been fortified by the veritable "religion of nature" whose priests and prophets included Schiller and Goethe in Germany, Wordsworth and Coleridge in England, and Thoreau and the new "transcendentalists" in America. Naturalism in this version was less an addition to the idea of progress than a deduction from it. Nor, in this version, was it incompatible with orthodox Christian teaching. What it took for granted was that every living thing had its place in Creation; that each more advanced form of life carried its primitive past with it; that in the dark struggles of the living world, beauty and order arise out of chaos; that man is not removed from these natural laws and forces; and that every production of nature is but a stage in the endless march of progress.

The naturalistic point of view is evident in a wide range of early nineteenth-century activities—in art, politics, religion, and science. In all of these endeavors, the leading figures employed what was called "the natural history method," a phrase found often in the books and essays of the period. What was understood by this was the firm connection between what a thing is and how it got to be that way. To understand civilization, one studies "savages." To understand government, one examines historically earlier forms of it. And, alas, to understand human nature, one consults the rest of nature. Note that one of Darwin's engaging works was his essay on the development of his own son. Not to make too much of this, it is enough to say that Darwin's studies of fossils and his careful observations of an infant son proceeded from the same naturalistic perspective that animated the artistic, historical, and philosophical creations of the nineteenth century.

The influence of this perspective is abundantly clear in the psychological thought of the period, and it is as clear before Darwin's great contribution as it is thereafter. Throughout the early decades of the nineteenth century, the general periodicals of the day were filled with accounts of life and habits of "the lower orders"—birds, bees, ants (especially), and fish. In this same period, the combination of philanthropic sentiment and the naturalistic perspective was responsible for many essays devoted to the elimination of cruelty in the treatment of domesticated animals. The arguments advanced in these essays were strengthened by constant and exaggerated reference to the *human* qualities possessed by cats, dogs, horses, and the rest.

The theory of evolution, as set forth by Darwin, was in this respect the crowning achievement of naturalism, not its earliest bible. It put in order and on a firm foundation of fact views that had already come to dominate the thinking of the age. Indeed, when we consult critical reviews of *Origin of Species* (1859) written by Darwin's contemporaries and published in the most influential journals of the day, we are impressed by the praise and general agreement displayed toward the work.[3] It should be recalled that the famous wave of anti-Darwinian rhetoric did not begin with *Origin of Species* but with *Descent of Man* (1871). The negative response was to Darwin's "metaphysics," not his naturalism.

The first writer of note to apply evolutionary theory to a broad spectrum of psychological issues was George Romanes (1848–1894) whose first large-scale work on the subject was *Animal Intelligence*[4] (1882). We would be inclined to judge this work today as coming under the heading of the ethology rather than comparative psychology, chiefly because it is utterly lacking in experimental content and method. Carrying on a long tradition—and one to which Darwin himself was wed—Romanes presents *anecdotal* information on the mental life of lower organisms. In his Preface to this work, he is careful to observe the limitations of the "anecdotal method," and insists that he will only include those observations made by highly regarded naturalists whose objectivity is beyond question. He also anticipates those critics who will challenge the view that animals have any mind at all:

> Does the organism learn to make new adjustments, or to modify old ones, in accordance with the results of its own experience? If it does so, the fact cannot be due merely to reflex action . . . Of course to the sceptic this criterion may appear unsatisfactory, since it depends not on direct knowledge, but on inference. Here, however, it seems enough to point out . . . that it is the best criterion available; and further, that scepticism of this kind is logically bound to deny evidence of mind, not only in the case of lower animals, but also in that of the higher, and even in that of men other than the sceptic himself.[5]

What Romanes insisted is that all attributions of *mind* are inferential, since we can be sure only of our own minds and not those of others. The basis upon which we infer that Smith has fear is that Smith is doing the sorts of things we do when we are fearful. The same is the case when we attribute to Smith the properties of consciousness, motivation, memory, etcetera. Romanes was aware of the fact that some of these attributions are less firm than others, and all of them become weaker as we observe organisms lower and lower in the evolutionary series:

> [A]s the dawn of unconsciousness or the rise of the mind-element is gradual and undefined, both in the animal kingdom and in the growing child, it is but necessary that in the early morning, as it were, of consciousness any distinction between the mental and the non-mental should be obscure . . .[6]

The thesis developed by Romanes is that consciousness is the ultimate stage of mental evolution, as the primitive reflexes are the first stage. Instincts occupy a middle ground. Although a celebrated biologist in his own right, he resists the temptation to defend this thesis with facts drawn from physiology. He notes that, as functions move from the level of reflex to that of instinct, and from instinct to the genuinely rational, "the nervous processes engaged are throughout the same in kind, and differ only in . . . their complexity."[7] Thus he opposes biological reductionism and urges that the thesis be evaluated at the level of observable behavior.

What we discover in Romanes' work is a transition from introspective psychology to behaviorism, with Romanes himself reflecting the assets and the deficiencies of each. What his argument amounts to is this: (1) I begin by consulting the facts of my own consciousness, recognizing the factors that led me to certain feelings, actions, and thoughts. (2) I then take these introspectively yielded data and match them up with the behavior of lower organisms. I make careful distinctions between genuinely adaptive and learned behavior and the more rudimentary instincts and reflexes. (3) When I observe animals doing the sorts of things which I do as a result of my psychological processes, I infer that they too possess similar processes. Thus, Romanes concludes that ants practice slavery, that the termite queen summons an audience, and that the trap-door spider learns how to prevent illegal entries! What is *behavioristic* in this approach is the commitment to accept only observable behavior as the evidence of psychological functions. What is *mentalistic,* of course, is the commitment to translate this evidence into the language of consciousness. Together, these aspects of Romanes' psychology are illustrative of anthropomorphism, a very serious charge by modern lights. What this *ism* is said to be guilty of is unwarranted inference and the installation of the human mind as the standard by which to explain the behavior of nonhuman animals. Thus we find one of Romanes' contemporaries, the famous botanist William Lauder Lindsay, devoting two very large volumes to the phenomena of mental health and mental disease in animals, and urging the formation of mental hospitals for them.[8]

The antidote for Romanes' version of anthropomorphism was soon provided by his admiring critic, C. Lloyd Morgan in his *Introduction to Comparative Psychology* (1894).[9] It was in his work that Morgan developed his famous canon:

> In no case may we interpret an action as the outcome of the exercise of a higher psychic faculty, if it can be interpreted as the outcome of the exercise of one which stands lower in the psychological scale. (53)

In this same work, however, Morgan takes for granted that "It is with . . . states of consciousness that psychology has to deal" (25), and we therefore may place him within the mentalistic context of nineteenth-century psychology. Unlike

Romanes, however, he was more impressed by the potential usefulness of physiology to the study of psychological processes and by the need to rid scientific psychology of the consequences of its introspective language. We see both of these concerns in the following passages:

> [I]f we accept evolution as the true basis of explanation alike in biology and in psychology, we are justified in inferring that . . . concurrent with the community of nervous mechanism and its physiological functioning, there is a community of psychical nature and psychological functioning. (84)

> One of the greatest difficulties against which the student of zoological psychology has to contend is, that the language in which he needs must describe and endeavor to explain the mental processes of animals embodies the results of a vast amount of analytical thought. He has to employ phrases which imply analysis, to describe experiences which involve no analysis. (87)

In the first of these passages, Morgan offers anatomy and physiology as the grounds on which inferences may be plausible. That is, we only infer a psychological similarity between two animals when, in addition to the observable behavior, there is a similarity in their neurological apparatus. In the second passage, he further refines his cautions. The scientist who observes ants or bees or dogs brings to these observations a language rich in analytical content. To him, two colliding masses of ants are "armies," because only armies of human beings behave the same way. And a similar observation of human armies would lead the observer to conclude that the campaign was directed by a certain "strategy" and toward a certain "goal"; that actions of a given type were "heroic" and others "cowardly." Yet, these terms, "which imply analysis," originate in the observer and may be wrongly applied to events in which no such analysis is present. Indeed, even in human affairs many occurrences can be explained without any reference to mind or consciousness. This is not to say that human beings lack either, but that a scientific psychology may confine itself to the observed behavior and to the neurophysiological processes by which it comes about.

So far, the picture that emerges from this review of Morgan's comparative psychology is one of an utterly modern psychologist, behavioristically inclined, and confident that the theory of evolution and the science of physiology will give psychology all it needs to attain scientific status. But toward the end of his text he offers this:

> To the question, Is mental development in all its phases entirely, or even mainly, dependent on natural selection through elimination? I reply with an emphatic *no* . . . I see no evidence to show that com-

> manding intellect, mathematical or scientific ability, artistic genius or lofty moral ideas, are attributable solely to natural selection. (355–356)

In this passage there is that element of conservatism found among a number of otherwise loyal Darwinists. Alfred Russell Wallace, for example, the cofounder of the theory of evolution, was also persuaded that an evolutionary account of ethics and of abstract artistic and mathematical creations simply fails.[10] It is only when we come to such radical evolutionists as Haeckel (1834–1919) that all such reservations are submerged in the triumph of monistic materialism. Thus, when Haeckel turns to those who would exempt art or ethics or sublime thought from the reach of evolutionary theory, he urges them to surrender their superstitions. As for the alleged exceptions:

> They merely cease to pose as truths in the realm of pure science. As imaginative creations, they retain a certain value in the world of poetry. . . . Just as we derive artistic and ethical inspiration from the legends of antiquity . . . so we will continue to do in regard to the stories of the Christian mythology.[11]

But this was merely polemical Darwinism, and the builders of modern psychology wisely avoided such material. Instead, they took from the theory of evolution what was most serviceable, and from Romanes and Morgan what was most in keeping with the tone and objectives of twentieth-century psychology. The first clear result of this was the Functionalist school discussed briefly in the next chapter. One of its major architects was James Rowland Angell (1869–1949), who would look back at the heat of the past and observe in the cool and confident language of the modern psychologist,

> Our whole tendency now-a-days is to recognize and frankly admit, that inasmuch as we must infer the psychic operations of animals wholly in terms of their behavior, we are under peculiar obligation to interpret their activities in the most conservative possible way.[12]

Angell had been William James' student at Harvard and John B. Watson's teacher at Chicago. Even in this brief passage, we can see the old mentalism and introspectionism receding, and the new objective behaviorism on the horizon.

Defining Psychology: Natural Science, Mental Science, Social Science?

Method v. what's psych subject matter?

The roots of modern neuropsychology are most firmly planted not in the purely speculative materialism of the eighteenth century, but in the clinical and experimental discoveries of the nineteenth. The contributions of Gall, Bell, Ma-

Start applying psych to social world

gendie, and others have already been cited (Chapter 10) in this connection. But it was only later in the nineteenth century that genuinely modern integrations of clinical and experimental findings were achieved. It is appropriate here to compare the old and the new views of mind and of mental illness within the general context of medicine and biology. A very useful basis for comparison is found in the history of law as it pertains to the "insanity" defense.

During the peaks of their ancient civilizations, both Greece and Rome wrestled with the question of legal responsibility and liability, and the laws of both made provision for diminished responsibility. Children, for example, were accorded special exemptions. But in dealing with adults, the presumption of personal responsibility was the rule, not the exception. In their developed form, the laws of Greece and Rome specifically outlawed the *vendetta* and the so-called "law of revenge" (*lex talionis*), and did so on the grounds that only the actor can be judged as responsible for the action in question. But what, then, of insane persons? Again, both Greek and Roman law acknowledged conditions of insanity and both granted a degree of relief to criminals judged to be insane. But the criterion of insanity was rather sharply set. In Rome, for example, the defendant had to be found to be *fanaticus* and *non compos mentis*. The former term is the equivalent of "wild," the latter, "no power of mind" or "no controlling mental power." In other words, the defendant had to qualify as something less than a human being.

This ancient "wild beast" standard was retained in Western law until the very beginning of the nineteenth century.[13] However, in the landmark British case of *Hadfield* (1800), this standard was directly and successfully challenged. Counsel for the defense argued that Hadfield, who was charged with treason (for attempting to take the life of George III), was laboring under a delusion, and that this delusion was very likely the result of brain injuries sustained in war.[14] Hadfield's acquittal illustrates the general willingness to accept neurological opinions as grounds for exoneration. We see, then, that as early as 1800 the "brain theory" of mind was sufficiently compelling to be decisive in a celebrated legal case. The point here is not that by 1800 even the man in the street had come to adopt a materialistic psychology, but that he was willing to concede that certain peculiar states of mind could be induced by pathological states of brain. Thus, the concept of free will was not abandoned, but was somewhat casually absorbed into the larger naturalistic framework. Hadfield, the reasoning went, did not act freely because his will was clouded by a delusion, and the delusion was conditioned by a diseased brain. Note that no such evidence was actually presented in this case; it was merely inferred. And note, too, that in the vast majority of similar cases, right up to our own day, the allegation of "brain disease" rarely is corroborated by clinical neurology in cases involving criminal offenses.

Within the more or less official councils of thought, however, there was (and is) a lingering debate on the extent to which mental life might be understood in purely neurological terms and by purely neurological methods. Throughout the nineteenth century, eminent spokesmen for both sides appeared and served up a wide assortment of arguments. Any brief attempt to summarize these runs the risk of libel, but we can at least strive to capture the essence of the competing views. On one account—let us call it the "mentalistic" account—the subject matter of psychology is consciousness and the best (if not the only) method by which it may be studied is the psychological method of self-examination. We call this approach "introspectionism," but it is important to recognize the subtler meaning of the term. As exercised by the earlier philosophical psychologists (for example Locke, Berkeley, and Hume), the introspective method was confined to an examination of one's own ideas and experiences on the assumption that all healthy persons had ideas and experiences in much the same way and according to the same basic principles. But with the advent of research on sensation, that is, actual experiments on perceptual processes, the introspective method was, as it were, externalized. Thus, the introspective method criticized by Wundt was the method of private or personal introspection of the philosophers. The method of *experimental* introspection, however, is nothing but the psychophysical methods developed by Fechner and praised by the same Wundt. Fundamentally, these methods proceed from the conviction that only a person having an experience can report it. Therefore, the proper study of such experiences (sensations, percepts, cognitions) necessarily depends upon objective observation and measurement of the self-examinations of the subject.

John Stuart Mill dubbed this the "psychological method" and he defended it against the utterly biologized psychology that Comte's *Positive Philosophy* demanded. Indeed, one of the reasons Mill had become disaffected with Comtean positivism—which he had stoutly supported earlier—was his conclusion that the positivist approach to a science of the mind was naive and constricted. The tension between Comte and Mill on this point was not a result of differing views regarding experimental science, for on this they were in complete accord. Instead, it arose out of differing views on the nature of explanation. Mill's defense of the "psychological method" was in keeping with his general and developed views on inductive science, his resistance to metaphysical speculation, and his disavowel of deductive methods when applied to matters of fact.

Let us now examine the other view, which, for the present purposes, will be called "reductive materialism." Its defenders begin with the proposition that all mental states, events, and processes originate in the states, events, and processes of the body and, more specifically, of the brain. To develop a true science of the mind, therefore, requires an understanding of the laws governing the organization and activities of the brain. For this, nothing more is needed than a careful examination of relevant clinical patients and a systematic program of research into the functions of the brains of the advanced species.

Language [illegible] is math.
(if can't reduce data to math, [illegible] data)

One of the most influential texts to arise out of this perspective (and to do so much to make it "official") was Henry Maudsley's *Physiology and Pathology of the Mind* (1867).[15] Maudsley is now known for many contributions. He was largely responsible for the creation of out-patient facilities in mental hospitals and for his powerful defense of the "medical model" of mental illness.[16] In the book of 1867, Maudsley paused to examine Mill's position:

> Mr. J. S. Mill has made a powerful defence of the so-called Psychological Method. In his criticism of Comte in the *Westminister Review* for April 1865, and in his "examination of Sir William Hamilton's Philosophy", he has said all that can be said in favour of the Psychological Method, and has done what could be done to disparage the Physiological Method. . . . [T]he admirers of Mr. Mill cannot but experience regret to see him serving with so much zeal on what seems so desperately forlorn a hope. Physiology seems never to have been a favourite study with Mr. Mill. . . . The wonder is, however, that he who has done so much to expound the system of Comte, and to strengthen and complete it, should on this question take leave of it entirely, and follow and laud a method of research which is so directly opposed to the method of positive science.[17]

In calling this perspective "reductive materialism," care must be taken not to confuse it with a new version of atomism or epicureanism or sensationism. The "reductive" feature refers to the requirement that the phenomena of mental life be reduced to the laws of neural function. Moreover, stress is laid to the dynamic nature of brain-function—its evolutionary character. Maudsley, for example, was as opposed to Mill's associationistic psychology as to his "psychological method":

> Infinite mischief and confusion have been caused by the habit of speaking of ideas as if they were the mechanical stamps of impressions on the memory, instead of as, what they truly are, organic evolutions in respondence to definite stimuli; our mental life is not a copy but an idealization of nature, in accordance with fundamental laws.[18]

Nor are we any longer confronting the odd "vibratiuncles" of a Hartley or the passive but "sentient statue" of a Condillac. The new neuropsychology is prepared to assimilate the entire range of psychological determinants as these come to affect the brain:

> When we are told that a man has become deranged from anxiety or grief, we have learned very little if we rest content with that. How does it happen that another man, subjected to an exactly similar cause

> of grief, does not go mad? It is certain that the entire causes cannot be the same where the effects are so different; and what we want to have laid bare is the conspiracy of conditions, internal and external, by which a mental shock, inoperative in one case, has had such serious consequences in another. A complete biographical account of the individual, not neglecting the consideration of his hereditary antecedents, would alone suffice to set forth distinctly the causation of his insanity.[19]

In contrasting the positions of Mill and Maudsley, we catch glimpses of an issue that pervaded psychology throughout the nineteenth century, whether that psychology was taking place in the laboratory, in the lecture hall, in the clinic, or in the psychiatrist's private office. Together, naturalism, Darwinism, and the idea of progress had the effect of challenging that traditional confidence of the arm-chair psychologists. No one any longer believed that the full set of psychological laws could be discovered by a Lockean or Humean form of speculation. Nor was there the same willingness to dismiss "madmen" on the grounds that a science need not waste time on accidents and eccentricities. Each individual life was now judged to be a *life in progress,* an *evolved* life that could only be understood and explained by charting its unique experiences, its unique heredity, and its dynamic states of consciousness.

But even in this, the nineteenth century was caught up in one of those titanic contradictions for which the Victorian age is ungenerously remembered. On the other hand, there was that evolutionary naturalism concerned with the history and the destiny of the species as a whole. On the other hand, there was that libertarian ethic that nearly spent itself in striving to secure the freedom and defend the dignity of every individual. The ways invented by that century to eliminate the contradiction were elaborate, confused, confusing, and are very much with us now. As suggested in the previous chapter, Wundt's approach was to insist on two distinct sciences: one addressed to the individual mind (its "manifold of consciousness") and relying primarily on the methods of psychophysics, the other addressed to social aggregates and relying on the methods of historical and anthropological analysis.[20] The former science would be quite naturally tied to other natural sciences, and especially to biology. Wundt makes it quite clear in his *Lectures on Human and Animal Psychology,* however, that neither radical materialism nor radical idealism will have a place in this science.[21] The focus remains ever on the facts of *consciousness,* and not on the alleged neural or spiritual causes.

> If we could see every wheel in the physical mechanism whose working the mental processes are accompanying, we should still find no more than a chain of movements showing no trace whatsoever of their significance for mind. . . . [A]ll that is valuable in our mental life still falls to the physical side.[22]

But Wundt's psychology of the individual mind is still "physiological" in the sense discussed in the previous chapter. It is a psychology of laws and principles which can be unearthed through the enlarged methods of psychophysics.

But on the matter of the second science, the "folk psychology," Wundt provides a most revealing side of the nineteenth century's struggle with determinism. The question at issue has to do with social man, the man of action, and not the idealized man examined in the perception laboratory. Now, it is all too common in discussions of the history of psychology to find Wundt described as a "voluntarist," and to make no connection between this and the balance of his psychological inquiries. We begin to detect the connection, however, when we recognize the implications contained in the above quotation and in those provided in Chapter 10. Wundt's commitment was to the psychology of *consciousness,* a science of mind as mind. When he insists in the above passage that "all that is valuable in our mental life still falls to the psychical side" even after we have studied brain mechanisms exhaustively, he is simply stating his essential defense of the psychological method. Thus, what is rejected is not so much the thesis of radical materialism but the correlated claim that the methods of the radical materialist are even germane to a scientific psychology. Again, let us be cognizant of Wundt's genuine respect for and thorough knowledge of the methods and findings produced by the neural sciences of his day. What he would not be led to by them, however, was the all too complacent belief that the problems of psychology would be (or could be) solved by them.

What was the source of his resistance? Quite simply, it was the facts of consciousness, itself, which established that significant human actions proceed from volition, and that volition was not explainable in terms of neural events.

To appreciate this analysis, it is necessary to understand what Wundt meant by "volition." If all he had meant was "desire" or "motivation," we would simply counter his claim with the well-established facts that connect brain activity with motivational states. But for Wundt, a motive is a uniquely psychological entity, not to be confused with biological drives or emotional states. A motive is a *reason for acting:*

> When we have taken account of every one of the external reasons that go to determine actions, we still find the will undetermined. We must therefore term these external conditions not causes, but *motives,* of volition. And between a cause and a motive there is a very great difference. A cause necessarily produces its effect: not so a motive. . . . [S]ince all the immediate causes of voluntary action proceed from personality, we must look for the origin of volition in the inmost nature of personality—in *character.* Character is the *sole immediate cause* of voluntary actions.[23]

Again, Wundt was not an anti-determinist, but he was an anti-mechanist. When he refers to "character," he refers to the complex creation of biological organization, cultural influences, hereditary predispositions, and that matrix of beliefs, opinions, attitudes, and feelings that give a person a unique identity. To understand this person, one must invoke the methods of historical science, not physical science. For to understand the person's *character* is akin to understanding an entire culture, and the determinants of the consciousness of that culture.

Many of these points can be found in the *Principles of Psychology* (1890) and other writings of William James (1842–1910), perhaps the most important figure in the history of American psychology. In James' books and essays, as in Wundt's, we discover that a scientific psychology begins with the facts of consciousness and proceeds to the laws of their organization. We discover also a general dismissal of the view that such facts and laws are (or even are in principle) reducible to physical or biological processes. And, finally, we discover a special place given to the *fact* of human will and the peculiar autonomy it displays in a variety of settings, not the least of which is the religious.

In the abridged (1892) version of his two-volume classic, James begins his presentation of psychology with allegiance to the biological perspective, and offers a materialism as a plausible hypothesis: "The immediate condition of a state of consciousness is an activity of some sort in the cerebral hemispheres."[24] He takes this hypothesis to be soundly supported by many medical and experimental findings and assumes therefore, "without scruple . . . that the uniform correlation of brain-states with mind-states is a law of nature."[25] His subsequent examination of the sensory and motor functions only tends to support this "working hypothesis." But then, when discussion turns to the matter of consciousness itself, James' loyalty to the hypothesis sustains first embarrassment and then abandonment:

> When psychology is treated as a natural science . . . "states of mind" are taken for granted, as data immediately given in experience; and the working hypothesis . . . is the mere empirical law that to the entire state of the brain at any moment one unique state of mind always "corresponds." This does very well till we begin to be metaphysical and ask ourselves just what we mean by such a word as "corresponds." This notion appears dark in the extreme. . . . [T]he difficulty with the problem of "correspondence" is not only that of solving it, it is that of even stating it in elementary terms. . . . We must know which sort of mental fact and which sort of cerebral fact are, so to speak, in immediate juxtaposition. We must find the minimal mental fact whose being reposes directly on the brain-fact. . . . Our own formula has escaped the . . . assumption of psychic atoms by *taking the entire thought* (even of a complex object) *as the minimum with which it deals on the mental side,* and the entire brain as the minimum on the physical side. But the "entire brain" is not a physical fact at

all! . . . Thus the real in physics seems to "correspond" to the unreal in physics, and *vice versa;* and our perplexity is extreme.[26]

James' friend, the distinguished philosopher C. S. Peirce (about whom more is said in the next chapter), had put the case more directly:

> The materialistic doctrine seems to me quite as repugnant to scientific logic as to common sense; since it requires us to suppose that a certain kind of mechanism will feel, which would be a hypothesis absolutely irreducible to reason—an ultimate, inexplicable regularity; while the only possible justification of any theory is that it should make things clear and reasonable.[27]

James, Wundt, Peirce, and their disciples shared an impatience with overly confident materialism, and an eagerness to rid psychology of the heavy metaphysical baggage that must be carried by those who insist that important questions are now settled once and for all. But these same attributes may be assigned to those like Maudsley who found in neurophysiology that very simplicity of description and economy of explanation for which philosophical psychologists had been searching for centuries.

The principle disputants in the free will-determinism controversy were drawn primarily from the quarters of "voluntarism" and "materialism," but the nineteenth century provided room for other perspectives as well. There were some observers who were convinced that the actual facts of human life were not properly comprehended or fully addressed by any of the prevailing doctrines; that introspection, confined to "states" of consciousness, was unable to discover the *laws* of mind; that Darwinism confused analogies with identities; and that materialism was hopelessly inadequate in the face of the complexity of social life. George Henry Lewes (1817–1878) is illustrative of the nineteenth-century thinkers who expected psychology to be a "positive" science, but who rejected the tidy *isms* promoted by their contemporaries. In his *Problems of Life and Mind: The Study of Psychology* (1874–1879),[28] Lewes argued more in the tradition of J. S. Mill and against those he judged to be uncritical in their acceptance of Darwinism, materialism, and introspectionism:

> The conditions of existence of mental phenomena are not only biological but also sociological studies. A serious investigation of these will serve to remove most if not all of the difficulties which make men cling to the spiritualist hypothesis, because they are profoundly impressed with the inadequacy of the materialist hypothesis. (Ch. IV, Sec. 61)
>
> We must quit Introspection for Observation. We must study the mind's operations in its expressions, as we study electrical operations

> in their effects. We must vary our observations of the actions of men and animals, by experiment, filling up the gaps of observations by hypothesis. (Ch. VI, Sec. 75)
>
> For Mr. Darwin's purpose it was needful that he should emphasise the position that "there is no fundamental difference between man and the higher animals in their mental faculties." . . . For our purpose it is needful to point out that, while there is no fundamental difference in the *functions* of the two, there is a manifest and fundamental difference in the evolved *faculties*. . . . When Comte affirms that there is nothing in Humanity, the germs of which are absent from Animality, the assertion requires qualification. Animals may be said to have the germs of our moral and intellectual life in somewhat the same sense that serpents have the rudiments of our limbs. (Ch. VIII, Sec. 101)

Note in these passages that Lewes is not so much denying determinism as he is rejecting mechanical versions of it and pointing to the wider social context within which determinative forces operate. Note, too, the insistence on the need for experimentation and the importance of objective observations of human and animal activity (behavior). Lewes does not oppose the study of animal psychology. He cautions, however, against overly "Darwinized" interpretations of such findings as may be produced by this study. In the end, the question of determinism is either meaningless or it admits of empirical test. In any case, the proper subject of psychology is the actual conduct of real persons in socially significant settings, their doings, their ideas, and the pressures operating on them.

As in the area of comparative psychology, so too in the nineteenth-century treatment of consciousness and mental life was psychology moving in the direction of the practical, the antimetaphysical, the antitheoretical. In some respects, the writings of Wundt, Lewes, James, Peirce, and the others diminished the general enthusiasm for a reductionistic physiological psychology. But the greater effect was to create a place for both psychology as a *natural* science and as a *mental* and *social* science. Thus, specialists did not suddenly appear, but certain methods and problems, which by their very nature came to be a collection of specialties, were gradually adopted.

The Darwinian perspective, too, tended to question the role of reductive materialism in psychology. With its emphasis upon dynamics, it opposed the traditional mechanistic theories of life and replaced them with a usefully general theory of functional adaptation. And with its emphasis upon progress through strife, it put the "life history" approach to psychological issues on a firm naturalistic foundation. More subtly, it imparted a pervasively pragmatic tone to psychological inquiry. It replaced the introspectionist's question, "What is this mental state?" with the functionalist's question, "What is it *for?*" And it tended to blur the age-old distinctions between instinct and rational choice, habit and purpose, accident and design. Again, Darwinism was rather the culmination than the origin of these positions.

Clinical Psychology and the Unconscious

The notion of unconscious sources of motivation is a very old one both in literature and philosophy. Even in the epics of Homer we discover dream-merchants sent by the gods to plant ideas and fears in the mind of the unsuspecting sleeper. So also is the more general notion that, underneath the veil of reason, our mental life actually proceeds according to the mandates of passion.

The history of theories of mental illness is a subject worthy of book-length treatment, but its essential character is revealed by two competing and conflicting assumptions about the very nature of the mind. One is, as we have seen in previous chapters, the naturalistic; the other is the spiritualistic. But in making this distinction, we are not always in a position to anticipate the corollaries that might be derived from these assumptions. Consider, for example, the "demonic possession" theories widely circulating from the late Middle Ages to the Elizabethan period, and the "lunar" theory of madness that replaced it. The argument for a demonic or spiritual possession of the mind has not always or often been based upon sheer superstition. Rather, it follows quite directly from a Cartesian-type theory of mind, a theory not inconsistent with Aristotle's account. Without carefully developing the argument, let us simply take the following propositions for granted: (1) The mind, in its highest level of activity, comprehends abstract principles, universals, and general concepts which are not given to or through the senses. (2) It is this rational faculty that allows rational beings to order their conduct in a purposeful manner and to bring their judgments into line with the realities of their lives and surroundings. (3) Since this rational faculty is in possession of knowledge that could not be gleaned from the senses, and since it pertains to that which is not material, the faculty itself cannot be a material faculty. (4) Since it is not a material faculty, it cannot be affected or influenced by anything that is purely material. Now, if all of this is accepted, it follows that the source of irrationality or genuine "insanity" must be immaterial, which is to say *spiritual*. That is, the only thing that can take possession of the rational faculty is that which is somehow like it and, as we have seen, the faculty in question is not like matter.

When we turn to the "lunar" theory, according to which states of madness are induced by the phases of the moon, we discover what was actually an early naturalistic theory of mental illness not unlike Mesmer's "magnetic" theory developed late in the eighteenth century. But note the grounds on which the rationalist (Aristotelian, Cartesian, etc.) would oppose such views. First, they entail the notion of "action at a distance" and offer no evidence of a medium

Repressed emotions related to conflicts, them may be real or fantasy. (not actual memories)
unconsious drives affect us

through which the moon or the magnet might reach the mind. But more importantly, they require that purely physical forces influence that which has no physical features. To the rationalist, therefore, lunar and magnetic theories are simply witchcraft by another name, and are steeped in the "natural magic" of Renaissance heremeticism.

Mesmer (1734–1815), it must be recalled, always insisted that the magnetic trances he induced were explicable in physical terms, and that there was nothing magical or superstitious about them. His own medical training had been the best (he completed his work at Vienna under such notables as van Swieten) and, until "Mesmerism," he had been a highly regarded physician. But when his claims and theories were officially reviewed they were dismissed on a variety of grounds, including the "action at a distance" problem.[29] Even later, when such prominent medical men as John Elliotson (1791–1868) and James Esdaile (1808–1859) tried to convey to their British colleagues the great anaesthetic power of "mesmerism," they all but lost their standing in professional circles. Elliotson found and edited *The Zoist* (1843–1856) precisely because the scientific journals of his day would not publish studies reporting the salutary effects of mesmerism.

This discussion strives to underscore how completely the scientific establishment had come to subscribe to a mechanistic materialism, and how resistant this establishment was to anything that smacked of the old "metaphysics." In England there was a particular disinclination toward theories of any kind, let alone those advanced by the mesmerists. This same disinclination was fortified by the excessive claims of the new Romantic idealists—the direct descendants of Schelling, Goethe, Fichte, and Hegel—who were now seeking to replace empiricism and natural science with the Absolute Idea. On their construal, the mere physical facts of the universe hide the grander purpose behind it and fail to reflect the central fact of freedom, toward which the human spirit is unconsciously but inexorably propelled. The scientists of the time not only did not weigh these propositions very carefully, they developed a keen sense for detecting signs of the Absolute in any new idea presented for their judgment, and were quick to reject it should the signs be there.

Against this background, it becomes easier to understand two aspects of the psychoanalytic theory propounded by Sigmund Freud: first, the cool reception it received in the medical circles of the day; and, secondly, Freud's eagerness to translate, wherever possible, the psychoanalytic concepts into neurological ones. Before returning to this, however, it is instructive to examine the broad features of Freudian theory.

Freud was the product of that marvelously contradictory climate of German thought in which science was defined in the positivisitc, deterministic, and physicalistic language of Helmholtz, and in which philosophy was Hegelian.

Free Association ...
(how mems are organized) to lead back to memories.
-sex + aggression (Darwinian)
.unconscious

Let us review the prevailing forces. Freud was born in 1856. He completed his doctorate (in neurology) in 1881. Wundt's laboratory had already been in existence and in highly productive existence for two years. Helmholtz, now sixty years old, was the senior statesman of German science. Evolutionary theory, which most of the leading antivitalist biologists adopted as fact, had already been in print for more than twenty years. Darwin himself was dead only six years. His most ardent and able disciples, Thomas Huxley (1825–1894) and Herbert Spencer, were both alive, and Spencer especially was promoting the psychology of "instincts." If we wonder how much of the new science was part of young Freud's education, we need only recall that his mentor in neurophysiology and neurology was Professor Ernst Brücke, himself a student of Müller's and one of Helmholtz's closest associates.

Philosophically the climate was overwhelmingly conditioned by Hegelianism. Psychology and phenomenology were two terms expressing the same subject: the subject of consciousness. The driving force in the universe was the free mind, a mind that had evolved from more primitive, nonreflecting, irrational substrates. Even Wundt, far from a Hegelian, had at least agreed that the subject matter of psychology was exhausted by the contents of consciousness. Wundt and Hegel also agreed on the central position of feeling in psychological life. If it is agreed that Freud had one of the great synthesizing minds of all time, we now have the list of major perspectives available for synthesis.

His energies as a senior graduate student were devoted chiefly to neurophysiological research under Brücke's direction. We know he hoped for a professional chair at the University of Vienna and that, as a Jew, his chances were negligible. The demands of marriage and family forced him into private practice in neurology, although, from the first, he never interrupted his research endeavors. Interestingly, he is cited in Wundt's *Principles* for a published article on aphasia that appeared in 1891.[30] A number of his patients suffered from "hysteria," which, at the time, was the diagnostic category invented to account for sensory and motor deficits occurring in the absence of detectable neuropathology. The French pathologist Jean Charcot (1825–1893) had revived and given respectability to the practice of hypnosis and was employing it in the treatment of certain hysterical cases as well as for anaesthetic purposes. Freud attended Charcot's lectures (1885–1886) at the University of Paris. He returned to Vienna, eager to apply the new technique, which had already caught the attention of Joseph Breuer, another of Brücke's former pupils. Freud would later write:

> Granted that it is a merit to have created psychoanalysis, it is not my merit. I was a student, busy with the passing of my last examinations, when another physician of Vienna, Dr. Joseph Breuer, made the first

The men remembered for great discoveries + theories always "stand on the backs of giants."

> application of . . . [hypnosis] . . . to the case of an hysterical girl (1880–1882).[31]

It was Breuer too who discovered, as one of his patients called it, the "talking cure" that later would receive the more impressive if less direct label "catharsis." While their medical relationship was not to last (though their friendship did), Freud and Breuer introduced the bare outlines of psychoanalytic theory in a series of jointly published papers beginning in 1895 and summarized in their *Studien über Hysterie.* Professor E. G. Boring gives a brief but searching insight into the theoretical bond between the two, Breuer being fourteen years the senior:

Ha!

> Breuer held that a certain amount of the organism's energy goes into intracerebral excitation and that there is a tendency in the organism to hold this excitation at a constant level. Psychic activity increases the excitation, discharging the energy. . . . What Breuer and Freud had from it, however, was the conception of psychic events' depending upon energy which is provided by the organism and which requires discharge when the level is too high. Because Brücke had trained them into being uncompromising physicalists, they slipped easily over from the brain to the mind.[32]

Under hypnosis, the patient's hysterical symptoms could be transferred from one part of the body to another. The hysterically paralyzed hand could be made to move, but with the consequence of immobilizing the intentional use of the leg. Sight could be restored, but with deafness following in its wake. Since the patient showed no understanding of the causes of the problem and since hypnotic induction could relieve the symptoms, Freud and Breuer concluded that the mechanism of symptom-formation was *unconscious.* Contrary to two widely held views, the symptoms were not feigned in an attempt to receive pity, nor were they limited to women (even though the term "Hysteria" derives from the Greek for *uterus*).

The unpopularity of Freud's notions, differences of opinion between him and Breuer, and Breuer's desire to quit science for practice, all worked to drive the two apart. By 1900 their relationship was almost exclusively social. It was in 1900 that Freud's *Interpretation of Dreams* was published. In the following year, *The Psychopathology of Everyday Life* appeared. Thus, as the twentieth century began, he had a following. By 1920 there was a movement. By 1930 he was, and has since remained, the preeminent figure in modern theoretical psychology.

Central to Freud's system is the concept of "unconscious motivation. Herbart and Fechner had both speculated about unconscious processes. But Freud's use of the concept was among the first to be scientific. That is, in the sense in which the term was introduced in the first chapter, a "scientific explanation" was generated by Freud's treatment of the idea. It became a "covering law" from which one could deduce certain consequences. His recourse to the concept was based on the need to discover a force or an agency by which behavior might be controlled independently of the patient's will. When Breuer's patient was relieved by the cathartic method, it was clear to Freud that the power of particular memories to control behavior was reduced once the actual memories were revived. Accordingly (and here the metaphor of the machine becomes reality), he reasoned that painful thoughts are *repressed,* that they come to reside in the unconscious, and that the symptoms are the work done by the repressed elements—that is, psychic energy is conserved through symptom-formation. Only through a conscious reliving of the former trauma, an exhumation of it from the unconscious reliving of the former trauma, an exhumation of it from the unconscious recesses, can the symptoms finally be removed. Short of this, they can only be moved about the body, disguised, accepted. Freud's own words could not be more descriptive:

Psycho energy

> *Hysterical patients suffer from reminiscenses.* Their symptoms are the remnants and the memory symbols of certain (traumatic) experiences.[33]

Freud's quick renunciation of hypnosis is to be understood not only in terms of his Helmholtzian bias—he described hypnosis as "fanciful" and "mystical"—but also as the result of his inability to bring a number of patients "under." He abandoned it completely in favor of the cathartic method and its complement, free association. Reasoning that neurotic symptoms are the result of incomplete or unsuccessful repression, Freud sought to discover the traumatic episode in the patient's life by a sort of subterfuge. This involved an analysis of so-called slips-of-the-tongue (*parapraxes*), automatic or free associations, and, most salient of all, dreams.

For Freud the determinist, there was no more reason to consider dreams and paraphraxes as uncaused than to consider normal speech or wakefulness uncaused. Thus, the "Freudian slip," no less than the disguised plots and fantasies of the dream world, under the penetrating lights of psychoanalysis, can be shown to have a direct bearing on the remnants of childhood traumas. These result from the very evolutionary progression that the human psyche undergoes on its way to an adult stage. Utterly consonant with Darwinism, Freudian theory rests on the notion of instinctual biological drives, that impel the individual to act in such a way as to survive. The principle governing or defining all these drives is that of *pleasure.* The exercise of this principle is sexual gratification which, in its most advanced expression, involves heterosexual relations for the express purpose of procreation. However, the individual arrives at this level only after successfully passing through more primitive stages of gratifi-

Sex drives everything we do

cation at any one of which the individual might be arrested by trauma. The stages are identified in terms of the particular source of pleasure: oral, anal, genital, and phallic. The sexual "energy" devoted to this endless search for pleasure is the "libido," which operates in the service of that most primitive and survivalistic element of psychological beings, the "id." Since the instincts of the id are such as to lead to incestuous, murderous, and purely egoistic actions, no human society could survive its unrestrained expression. Every society, then, must "socialize" its young: develop in them a conscience, or "superego," which will direct the individual not away from gratification but toward socially acceptable means of gratification. The compromise between the impulses of the id and the constraints of the superego results in the self of whom we are aware, the "ego."

On the way to mature sexual motivation (this being the motive to procreate through heterosexual encounters), the child must select appropriate objects. The male child's natural proclivity is for his mother, who has been abidingly associated with gratification. However, to court one's mother is, simultaneously, to court a showdown with one's father, the effect of which is or may be castration. An Oedipal complex results, in which the boy can only succeed in removing the threat of castration by becoming his father's equal, or by shifting his quest to another object. Pathological conditions result from the failure to resolve the tensions implicit in the Oedipal stage.

Even from this mere sketch of the Freudian theory of personality, we are able to appreciate the synthetic features of the system. The human personality *evolves.* Its origins are animalistic, survivalistic, Bentham's pleasure principle has been raised to the level of a medical or neurological reality. The engine of psychological growth is energy, which behaves according to the same sorts of laws prevailing in the physical world. As with physical energy, psychic energy is conserved, directed, and partitioned, but never destroyed. With Fichte and Schelling, Freud saw the world as a set of polarities, with the forces at one pole opposing those at the other, an id struggling for supremacy over a superego, a "life force" (*eros*) in heedless conflict with a "death wish" (*thanatos*), natural instincts driving the organism toward the next stage in the face of societal taboos and rites that might otherwise frustrate instinctual expression—and with, through all of this, Freud constantly asserting that the ultimate topic of psychological concern is consciousness in that daringly Hegelian sense: the knowing, feeling, mind in search of an Absolute that might bring it peace and harmony.

A group was soon to form around Freud and to create the Freudian "school" of psychoanalysis. Membership in it was a fluctuating affair, primarily because of Freud's insistence on strict orthodoxy and because of the pettiness of his reactions to disagreement. On the whole, the departures from orthodoxy were largely matters of emphasis rather than fundamental disagreements over the central terms of the theory. At least at the outset, those who might otherwise be called "Freudians" were distinguishable in terms of the relative importance they gave to social as opposed to instinctive determinants, and to where in the hierarchy of instincts they placed the several contained in Freud's theory. Carl Gustav Jung (1875–1961), for example, was more self-consciously "Darwinian" in his conviction that the unconscious contains not only the repressed elements of an individual's personal experiences but also the *collective* unconscious elements characteristic of the entire race. Thus, every male has an idealized or "archetypical" unconscious picture or conception of the female (*anima*), and every female has the corresponding male archetype (*animus*) within her unconscious. Accordingly, and in contrast with Freud's notions, the unconscious cannot be understood solely or even primarily through an analysis of personal history, nor can neurosis be understood solely or even primarily as the consequence of sexual forces.[34] Alfred Adler (1870–1937) departed from Freudian orthodoxy also and, like Jung, founded his own "analytical" school of psychoanalysis. Central to Adler's position is the notion of a "will to power," and the dynamic interaction taking place between an ever-evolving self and the shifting fortunes held out by society. Thus, in Adler we discover a movement away from Freudian determinism and its implicit dependence on physiology and the natural sciences.[35] Both the Jungian and the Adlerian alternatives were more formal breaks with Freudian orthodoxy, but long before either there were psychoanalytic perspectives quite at variance with the theory for which Freud would become so famous.

To pick up the non-Viennese lineage of psychoanalysis, we turn to France and the productive career of Pierre Janet (1859–1947), who received his doctorate under Charcot, whom he succeeded as head of the Psychological Laboratory at the Salpêtriere in 1890. Jung himself had gone to Paris to study with Janet, whose international repute was already great by 1906 when he was invited to give a course of lectures at Harvard. In addition to his own original thoughts on the nature and causes of mental illness, the value of examining Janet's work from a historical perspective is that it sheds additional light on the problems Freud was to have with the scientific community. In this connection, Janet's most original work is also one of the few to have been translated into English—"The Mental State of Hystericals"[36]—and we will briefly consider it here.

Janet's chief interest was the diverse phenomena of hysteria to which Charcot had already devoted his energies and to which Freud would soon devote his. Hysterical symptoms ran the gamut from sleep-walking and morbid fear to "automatic writing" and profound losses of sensitivity and locomotion. The darkness that surrounded these disorders is suggested by the very term derived from the Greek word for *uterus*. Indeed, until Charcot had documented a number of cases of what he called "virile hysteria," the medical consensus stood behind the claim that only women were afflicted by it.

Given the nature of hysterical symptoms, sufferers generally sought the opinion and treatment of neurologists. But as Charcot and Janet demonstrated so convincingly, genuinely hysterical patients displayed no signs whatever of neuropathology. Moreover, although the symptoms limited neurological disturbances, they did not behave like such disturbances when carefully observed. Hysterical anaesthesia is illustrative. When anaesthesia is the result of neurological disease, all the reflexes associated with the now-lost sensations are also eliminated. But in hysterical anaesthesia, the sensitivity is gone while the reflexes remain generally unaffected. Moreover, hysterical anaesthesias are highly mobile and variable, changing not only in severity but in their location. The patient may have no sensitivity in the upper extremities on Thursday, and then none in the lower extremities on Friday—at which time the upper extremities display normal sensitivity.

The scientific question that arises in the face of such phenomena concerns first the issue of method and then that of explanation. Janet at this point was intimately familiar with the research of Wundt and his colleagues at Leipzig, and was convinced that their methods were unsuitable. The general aloofness of the Leipzig group toward complex medical problems—their "purist" stance on the nature of science—caused them, in Janet's view, to suffer a self-imposed naiveté. Even their notion of elementary sensations is largely stripped of psychological content, since their focus is on the verb (I *see,* I *feel,* I *hear*) to the exclusion of the subject (*I*).

> [T]he words "I, me" . . . are terms enormously complex. This is the idea of personality; that is to say, the reunion of presentations, the remembrance of all past impressions, the imagination of future phenomena. It is the notion of my body, of my capacities, of my name, of my social position, of the part I play in the world; it is an ensemble of moral, political, religious thoughts, etc.; it is a world of ideas. . . .[37]

Thus, Janet will not adopt the experimental methods of Wundt, for what concerns Janet is a malady of the *personality* as it reveals itself in perception or action or emotion; not a malady of perception per se. But if the methods of introspection and psychophysics are insufficient, must not those of the physiologist be used if the entire enterprise is to remain within a scientific context? This, of course, was the central issue, and what would come to be called the "psychoanalytic" method was designed to settle it. In Janet's hands, the method proceeds, as he says in his Introduction, from a "strict determinism" which ignores the study of "philosophic problems," and which begins with an objective description of the patient "in and by himself."[38] The overarching determinism accepted by Janet dictates that every mental phenomenon originates in the brain and that all mental pathology is of cerebral origin. There is no inconsistency between this thesis and the fact that the hysterical symptoms are not neurological. The latter term generally suggests some disease in the sensory or motor pathways or some lesion in a region of the brain. Janet's position is that hysteria does not come from conditions of this sort, but from the overall functional features of the brain ~~as a whole~~. Specifically, he traces the symptoms to some "fixed idea" controlling the patient's mental life and so "narrowing the field of consciousness" as to render the patient inaccessible to external events. The paralyzed patient is unable to "represent" movement in the cerebral centers because of the uncoupling that has taken place between the personality and the real world.

In writing a brief Preface to the first (French) edition of Janet's text (1892), the great Charcot tells readers that his former student "wished to unite as completely as possible medical studies with philosophical studies," but in these comments Charcot was referring more to his own increasingly psychological perspective than Janet's. Throughout the text, Janet is critical of premature attempts at theory and of all philosophical approaches to the clinical problems with which he is dealing. Each patient must be treated uniquely and not according to some overly general theory of science of the ipsedixtis of the philosopher. When Janet requires the concept of the unconscious, he ties it to specific symptoms, to the observed behavior of the patient. We see this in his discussion of the fixed ideas:

> 1st. These ideas may develop completely during the attacks of hysteria and express themselves then by acts and words. 2nd. In dreams more or less agitated which take place during sleep and natural somnambulisms and which often happen unexpectedly; it is at such moments that fixed ideas are wholly confessed. 3rd. One of the best processes consists in causing the patient to enter artificially into a state similar to the preceding ones—namely an imposed somnambulism. . . . 4th [W]hen it is possible to employ it, the process, which consists of utilising automatic writing is more exact than all the others.[39]

But when Janet turns to the recent publication on hysteria by Freud and Breuer, he finds several deficiencies, even though the authors graciously acknowledge Janet's priority. His complaint with their article is that they attempt to account for all hysterias with explanations that actually apply only to certain cases and that they underestimate the complexities of therapy:

> We formerly showed that . . . one could not treat the hysterical . . . before having reached those deep layers of thought within which the fixed idea was concealed. We are happy to see to-day MM. ~~Breuer~~ and Freud express the same idea. It is necessary, they say, to make this provocative event self-conscious; bring it forth to the full light. The [symptoms] disappear when the subject realises those fixed ideas. We

do not believe that the cure is so easy as that. . . . The treatment is unfortunately of a much more delicate nature.[40]

At the time Janet was writing these words, Freud had not yet developed his psychosexual theory of neurosis, but sufficiently kindred ideas were abundant enough for Janet to address the allegedly erotic element in hysteria:

> This erotic disposition exists, we think, in hysteria just as all possible fixed ideas do. We have collected, out of a hundred and twenty observations, four where it plays a role altogether predominant. There is nothing very strange in this. Amorous passions, sexual desires, must exist with these persons, most of them young, as they do with others. Hystericals hear people talk of love, see the evidence of it, read descriptions of it;—why should their mind, so ready to receive impressions, so docile to all influences, resist this one. . . . [P]atients speak very often of lover, husband, rape, pregnancy, etc. Now, you cannot put into your dilirium what is not in your mind. These things greatly preoccupy people of that age and sometimes of another age. It is quite natural that they should manifest them in their dreams.[41]

We discover in these passages that very tentativeness and common-sensical tone that would be so lacking in Freud's books and essays. Where Janet is willing to take the facts as they come, Freud would be nearly fretful in his attempt to fit the facts to his theories. To his contemporaries, there was something entirely unscientific about his very manner of discourse but, to posterity, it was precisely this manner that attracted attention, loyalty, and even devotion. Freud often thought of World War I as a kind of laboratory in which his central claims received macabre validation. In certain aspects, it was this war that gave a morbid cogency to Freudian theory among startled and saddened intellectuals. What is clear, however, is that the chilly reception his ideas suffered initially is not to be explained in terms of Victorian proprieties. Sexuality had been part of the clinical literature for decades by the time Freud made it a fixture in his writings. Anthropologists, at least since late in the eighteenth century, had reported on various primitive lusts and taboos, and had noted the centrality of sexual symbolism in many cultures. And the theory of unconscious motivation, with its ancient and even biblical roots, had already received refinements from Charcot and Janet before Freud had begun to assemble a theory. Freud's main problem—and here we must demystify that portrait of Gothic struggle painted by his friends and friendly biographers—was that his way of thinking was perceived as anachronistic. This not only accounts for the early disregard paid to his works, but to his own life-long attempt to confer scientific respectability on them. The scientific community had learned to live without the Absolute, without metaphysics, and without those cosmic integrations of an earlier age at just about the time Freud's psychoanalytic ruminations were published. Psychology particularly was earnestly seeking its place in science and so was especially embarrassed by a theory in which the ratio of facts to assumptions was so small.

If the earliest audiences for Freudian psychology were small and somewhat hostile, posterity more than compensated. But "clinical psychology" is more than and is different from "Freudian psychology." At the outset, it was medical psychology and, until the short-term triumph of psychoanalytic theory, it remained medical psychology. A distinction must be made, however, between this specialty and both the older and the more current forms of physiological psychology. As the term is used here, medical psychology addressed itself primarily to the phenomena of mental illness and to the application of experimental procedures by which these might be explained. Physiological psychology, both in Wundt's sense and in the current sense, has been chiefly concerned with those law-governed processes that characterize normally behaving, normally perceiving organisms.

A further distinction may be drawn along national lines, for in significant respects it was France that took the lead in medical psychology, Germany in physiological psychology. To give substance to this distinction, it should be recalled that Henry Maudsley was not an experimentalist or a "scientist" in the usual sense, but a practicing physician engaged in the sorts of things ordinarily associated with psychiatry. Thus, to the extent that clinical psychology identified itself with the natural sciences, it was an experimentally oriented discipline presented as something of an alternative to the introspective and psychophysical schools of Germany. It took the phenomena of the diseased mind not simply as conditions to be treated, but as suggestive of the principles by which mind as such could be understood. Disease, after all, is one of nature's experiments and the keen observer can learn as much from experiments of this sort (if not more) than can be gathered from any number of (unnatural, antiseptic, etc.) Leipzig studies. This was a general attitude in French psychology, but the psychologist who spoke most convincingly in its behalf was Alfred Binet (1857–1911).

Binet was born in Nice and received his degree from the Sorbonne under Professor Beaunis, who, upon retirement, turned his laboratory over to Binet. In the course of his abbreviated career, Binet co-founded France's most prestigious psychological journal, *L'Année Psychologique,* wrote authoritative texts on hypnotism, multiple-personality, and experimental psychology, and inaugurated those methods of intelligence-testing which came to define the field of mental tests and measurement. He was a close friend to Pierre Janet, a correspondent with William James, a regular author in international journals, and perhaps the person most responsible for keeping clinical psychology within the natural science framework at a time when theory-spinning threatened it with expulsion.

The Binet best known to psychologists is the co-founder of the Binet-Simon test of mental abilities, the test that would surface in America as the "Stanford-Binet." The test itself had an interesting history. It came into being after the Education Ministry called for a means by which retarded children could be singled out for special education in the Paris school system (1904). In not too many years, the test prepared by Binet and Simon were in use throughout the world and were even obligatory in certain states in America.[42]

In attending too closely to this phase of Binet's work in psychology, however, the contemporary psychologist often neglects not only Binet's larger contribution but the very basis upon which he came to develop tests of intelligence. Binet's interest was in the dynamics of mind, its development, disorders, and practical functions. Even in his clinical application of hypnosis, the dominating motive was to employ hypnosis as an *experimental* means by which to induce states of mind. In one of his earlier essays, "Mental Imagery" (1892), he discusses hypnosis as a method of "intellectual and moral dissection"[43] permitting the insertion of thoughts and ideas in the mind without having to rely on the subject's perceptual and sensory functions. It also permits the scientist to create cognitive states in the normal mind rather than waiting for the interesting clinical case. In his widely read text *On Double Consciousness,* he complains that while the universities continue to teach "an antiquated science, whose only method is that of introspection," French psychology is firmly tied to the natural sciences and has "left the investigations of psychophysics to the Germans."[44]

When we turn to his work on mental testing, we discover the same underlying considerations. Both in *The Intelligence of Imbeciles*[45] and *The Development of Intelligence in Children,*[46] Binet speaks of mental tests as forming an integral part of the "psychogenetic method." His primary interest in mental retardation arose from the judgment that the retarded individual—no matter the chronological age—presented a mind that was fixed at a given stage of development and was thus characteristic of that stage. In other words, what makes retardation pathological is not that its processes are unlike those found in normal persons, but that they are found only in much younger persons. Accordingly, careful tests of the retarded mind provide a key to our understanding of the normal evolution (genesis) of mind per se. Nonetheless, Bient recognized the immense variability displayed by this evolution and the need for a scientific psychology to address the facts of individual differences. In his essay "Mental Imagery," we find him praising Francis Galton for the statistical techniques that permit comparisons of individuals. Toward the conclusion of the essay, Binet observes:

> The whole present tendency of psychological research is to show, not that the mental operations of all person are of a similar nature, but that immense psychological differences exist between different individuals.[47]

Implicit in this passage is a criticism of the entire epoch of philosophical psychology and its residual, the introspective method. Experimental hypnosis, on the contrary, permitted inquiries into the differences between minds and allowed the investigator to create, repeat, and reverse mental states. Similarly, tests of mental functioning constituted a kind of indirect experiment in which mental age was the dependent variable and chronological age the independent variable. In *The Intelligence of Imbeciles,* Binet underscores the role of abstract thought in all genuinely mental operations and draws attention to the *functional* nature of this thought. Unlike the introspectionists, who labor so tirelessly to isolate the structural elements and atoms of experience, he states,

> we oppose [the] counterpart, that which gives action as the end of thought and which seeks the very essence of thought in a system of actions.[48]

Through the efforts of Binet in France and William James in America, the emerging science of psychology was given a broad middle ground between the psychophysical investigations of Germany and the increasingly narrow physiological investigations of the early neuropsychologists. This broad middle ground was occupied often by attention to psychopathology, but—at least with Binet and James—more often by research and theory devoted to normal cognitive functions. The practical and "pragmatic" tone of their writing was not at the expense of the mental, but it was at least indirectly supportive of a behavioristic (action-oriented) science.

The specialties incubated by those discussed in this section are no longer appropriately placed under the heading, "Clinical Psychology." What unites the otherwise diverse theories and practices of Freud, Janet, Binet, James, Jung, and Adler is first a recognition of mind as an entity with a past and a future, a dynamic entity whose principles of operation are not likely to be discovered simply through an analysis (introspective or psychophysical) of short-term sensations. Secondly, all of them shared in the conviction—but on different grounds—that genuinely psychological phenomena had to be dealt with at the level of psychology rather than at the (largely unknown) level of neurophysiology. Janet and Freud both subscribed to a "cerebrogenic" theory of hysteria, but both insisted that the disease must be described and treated at the psychological level. As a group, then, they resisted both introspective and biological reductionism. Finally, all of them argued in behalf of a *useful* scientific psychology; one able to explain mental phenomena as these occur in the real world and to actual persons.

By today's standards, William James would scarcely qualify as a clinical psychologist, even though he was medically trained and wrote often on clinical cases. Binet, too, was an experienced clinician whose essays and books often focus on aspects of psychopathology, but Binet would no longer qualify as a clinical psychologist. In putting them under this heading, therefore, is the present chapter guilty of an abuse of language? I think not. Instead, I would point to the even greater degree of specialization that has taken place in the past half century, the tendency to follow only one of the lines of inquiry instead of the many pursued by each of these men in his own professional life. For Binet, however, it would have made no more sense for a psychologist only to administer tests of intelligence than for a physician only to record temperatures. Mental tests were conceived as a means, not as the end, of scientific activity. So also with Janet, Freud, and Jung: The aim of psychoanalysis is not simply (if laudably) to cure a patient, but to come to understand the laws of mental life.

What we have seen in this chapter is the further refinement of general notions passed on late in the eighteenth and early in the nineteenth centuries. The refinement led to specialization both in the choice of problems and the choice of methods. Between 1870 and 1920, the distinct fields of personality, clinical psychology, comparative psychology, physiological psychology, and developmental psychology were all established. So, too, was that sociological point of view urged by Lewes, exercised by Wundt, and pervasive in the scholarship of the late Victorian period—the point of view that would soon become formalized as social psychology. We have, then, the bridge between the somewhat loose and hopeful "scientific psychology" and the more familiar fields and schools of contemporary psychology. What is missing from the record, however, is that further distillation that imparts the very special flavor of today's psychology. In the next chapter, therefore, we will examine three representative—dominant—orientations in modern psychology: behavioristic, Gestalt, and physiological. In Chapter 10, the nineteenth century was viewed through a wide-angle lens; in Chapter 11, more narrowly and in its latest stages. In the final chapter, we will be concerned with a briefer span, roughly 1920 to 1950, in which today's psychology was given its most immediate direction.

The Nineteenth Century's Invention

It was Alfred North Whitehead who credited the nineteenth century with having "invented the method of invention." It was the first period in history in which very nearly every important philosophical figure recognized the essential function of experimentation in the search for truth. It was also the first century in which the majority, the vast majority, of serious works in science were completely devoid of theological colorations.

The record of the century is particularly commendable in regard to psychology. When we examine the topics now filling the literature in professional psychology, we are hard pressed to find one that was not put forth—often in a form still to be improved upon—by those whose efforts we have examined in this chapter. Physiological psychology, little more than the product of polemicism in the eighteenth century, became a science in the hands of Flourens, Gall, Bell, Magendie, Helmholtz, and Wundt. Comparative psychology was invented by Spencer, Darwin, Romanes, and Morgan. The psychology of individual differences is the creation of Binet and Francis Galton, as are several of the statistical procedures needed for such studies. Cognitive and Gestalt psychologies are so intimately tied to phenomenology that only a purist could deny Hegel and the neo-Hegelians the title of "founders." "Freud," "Janet," "Jung," and "the unconscious" are near-synonyms. Our sense of what an experimental science is and ought to be is taken over, with only the slightest modifications, from J. S. Mill, and the general attitude toward the status of science remains largely the one advocated by Auguste Comte and his positivist disciples. Our fascination with hedonistic ethics, with the possibility of shaping the world through the processes of reward and punishment, is linearly traceable to Jeremy Bentham and the utilitarian movement. Even our vaunted "humanistic" psychologies, with their focus on "self-actualization," personal growth, and individual freedom, have never improved upon the original formulations by the German Romantics. Contemporary psychology then is largely a footnote to the nineteenth century.

Notes

1. Consult the entry "Rorarius" in Bayle's *Dictionary.* It can be found in *Historical and Critical Dictionary* (Selections), translated by Richard H. Popkin, Bobbs-Merrill, Indianapolis, 1965.

2. A six-volume English tradition of Gall's *On the Functions of the Brain and Each of Its Parts* was published in Boston in 1835. I have collapsed these, without abridgement, into Vols. XVI, XVII, and XVIII (series A) of *Significant Contributions to the History of Psychology,* edited by D. N. Robinson, University Publications of America, Washington, D.C., 1978. In my separate Prefaces to each of these volumes, I attempt to locate Gall's priority and influence in the areas of physiological psychology, comparative neuroanatomy, and fetal and neonatal anatomy.

3. Representative reviews are reprinted in "Darwinism," in *Significant Contributions to the History of Psychology,* Series D, Vol. IV.

4. Romanes, George, *Animal Intelligence.* The edition of 1883 is reprinted in *Significant Contributions to the History of Psychology,* Series A, Vol. VII.

5. Ibid., pp. 4–6.

6. Ibid., p. 13.

7. Ibid., pp. 12–13.

8. William L. Lindsay, *Mind in the Lower Animals in Health and Disease,* 2 vols., London, 1879. The two volumes are reprinted in *Significant Contributions to the History of Psychology,* Series D, Vols. VI, VII.

9. C. Lloyd, Morgan, *Introduction to Comparative Psychology,* London, 1894. This work is reprinted in *Significant Contributions to the History of Psychology,* Series D, Vol. II.

10. See especially the last chapter (XV) of his *Darwinism: An Exposition of the Theory of Natural Selection with Some of Its Applications,* Macmillan, London, 1889. In that chapter, Wallace argues that "It is impossible to trace any connection between (the musical faculty) and survival in the struggle for existence," and offers "Independent Proof that the Mathematical, Musical, and Artistic Faculties have not been Developed under the Law of Natural Selection" (pp. 468–469 of the edition of 1897).

11. This is taken from Ernst Haeckel's *Last Words on Evolution,* translated by Joseph McCabe, London, 1906. Reprinted in *Significant Contributions to the History of Psychology,* Series D, Vol. III, p. 111.

12. Angell's essay appeared as a journal article ("The Influence of Darwin on Psychology") in 1909, and is reprinted in *Significant Contributions to the History of Psychology,* Series D, Vol. IV. The quoted passage is on page 169.

13. For a brief review of this history, consult Ch. 2 of my *Psychology and Law: Can Justice Survive in the Social Sciences,* Oxford, New York, 1980.

14. This case of *Hadfield* and related British cases are given in Series F ("Insanity and Jurisprudence"), Vol. VI, of *Significant Contributions to the History of Psychology.*

15. Maudsley's text is reprinted in Series C, Vol. IV, of *Significant Contributions to the History of Psychology.*

16. For this defense, see his *Responsibility in Mental Disease* (1876) which, with Phillipe Pinel's *Treatise on Insanity* (London, 1806) is reprinted in Series C, Vol. III, of *Significant Contributions to the History of Psychology.*

17. Ibid., p. 37.

18. Ibid.,p. 187.

19. Ibid., p. 197.

20. The unabridged two-volume translation of Wundt's *Elements of Folk Psychology: Outlines of a Psychological History of the Development of Mankind,* London, 1916, is reprinted in Series A, Vol. XV, of *Significant Contributions to the History of Psychology.*

21. W. Wundt, *Lectures on Human and Animal Psychology,* translated by Janet Creighton, London, 1894. Reprinted in Series D, Vol. I, of *Significant Contributions to the History of Psychology.*

22. Ibid., p. 446.

23. Ibid., pp. 432–433.

24. W. James, *Text Book of Psychology,* Macmillan, New York, 1892, p. 5.

25. Ibid., p. 6.

26. Ibid., pp. 462–464, abridged.

27. C. S. Peirce, "The Architecture of Theories," *The Monist,* January 1891, pp. 161–176.

28. The separate work, *The Study of Psychology,* was published in London (1879) and is reprinted in Series A., Vol. VI, of *Significant Contributions to the History of Psychology.*

29. The assembled committee of reviewers included the aging Benjamin Franklin, from whom Mesmer had expected a more tolerant and informed hearing.

30. Wundt, *Lectures,* p. 308. Here, in a footnote, Wundt refers to Freud's "Zur Auffassung der Aphasien" (1891).

31. S. Freud, "The Origins and Development of Psychoanalysis," *American Journal of Psychology,* 21 (1910).

32. E. G. Boring, *A History of Experimental Psychology,* Appleton-Century-Crofts, New York, 1951, p. 709.

33. Freud, *Origins.*

34. The readiest source of these views is *Psychological Reflections,* selected and edited by J. Jacobi, New York, 1953.

35. Two representative works are *Inferiority and Its Psychical Compensation* (translated by S. E. Jellife, New York, 1917), and *Practice and Theory of Individual Psychology* (translated by P. Radin, New York, 1927). Adler's rejection of (Freudian) determinism is most clearly set forth in his *The Neurotic Constitution,* (1912).

36. P. Janet, *The Mental State of Hystericals,* translated by Caroline Corson, New York, 1901. This translation is reprinted in Series C, Vol. II, of *Significant Contributions to the History of Psychology.*

37. Ibid., p. 35.

38. Ibid., p. xiv.

39. Ibid., pp. 280–281.

40. Ibid., p. 412.

41. Ibid., p. 215.

42. It is one of the ironies of history that these tests, which sixty years ago were made mandatory in the interest of fairness and impartiality, are now under legislative and judicial attack.

43. The essay "Mental Imagery" was initially an article in *Fortnightly Review* (1892) and has been reprinted in Series B, Vol. IV, of *Significant Contributions to the History of Psychology.*

44. Binet's *Alterations of Personality* (English edition of 1896) and his *On Double Consciousness* (English edition of 1890) are given together in Series C, Vol. V, of *Significant Contributions to the History of Psychology.* The quotation is taken from p. 12 of the latter.

45. A. Binet, *The Intelligence of Imbeciles* (English translation, 1909). This has been reprinted in Series B, Vol. IV, of *Significant Contributions to the History of Psychology.*

46. A. Binet and T. Simon, *The Development of Intelligence in Children* (English translation, 1911). This is reprinted in Series B, Vol. IV, of *Significant Contributions to the History of Psychology.*

47. Ibid., p. 104.

48. Ibid., pp. 153–154.

Operant – consequences
Classical – contiguity.

12 Contemporary Formulations

Behavior → seeks to be science as (most [illegible]) defined by J. S. Mill.

Method as Metaphysic

To describe contemporary psychology with the abrasive term "footnote" is to tilt with the error of *scholastica successionis civitatium,* which we pledged to avoid in the very first chapter. No contemporary psychologist diffidently searches the annals of nineteenth-century scholarship in order to discover the problems or methods appropriate to psychology. The most casual inspection of the courses and the texts in psychology, at both the undergraduate and the professional levels, will prove beyond doubt, if not beyond concern, that very little of the debt to the nineteenth century is consciously acknowledged. The aspiring psychologist might be expected to know something about "Mill's methods" and that Wundt founded the first laboratory devoted exclusively to psychological research. It is also the expectation of a faculty that its students will be conversant with the general features of Darwinian biology and the sensory-physiological theories Helmholtz. But no one is asked any longer to pore over the works of Bain and Spencer, Fichte or Schelling, Kant or Hegel. Even William James is presented as a museum piece, and E. L. Thorndike as the author of a law and the prophet of a method that have both changed beyond recognition. No introductory course in psychology and certainly no concentration or "major" in psychological studies is considered complete or even respectable unless the great old names are periodically trotted out, dusted off, congratulated for having seen farther than most, and then gently returned to their crypts as psychology gets on with more pressing matters. Usually, as anyone willing to take the time to glance through the more popular general and historical texts will see, the list of great old names is expeditiously contracted. "The Greeks" of course, are never neglected. Perhaps a few lines will be devoted to the fact (except it isn't a fact) that not very much of intellectual consequence took place from the fall of Rome until the Renaissance except, perhaps, for St. Augustine